

5.3 Transformations of Sine and Cosine Worksheet #2

MCR3U

1) A sinusoidal function has an amplitude of 5 units, a period of 120° , and a maximum at $(0, 3)$.

a) Represent the function with an equation using a sine function

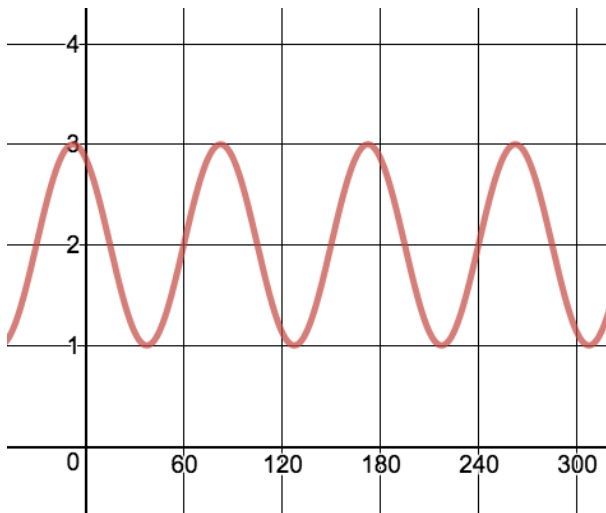
b) Represent the function with an equation using a cosine function

2) A sinusoidal function has an amplitude of $\frac{1}{2}$ units, a period of 720° , and a maximum at $(0, \frac{3}{2})$.

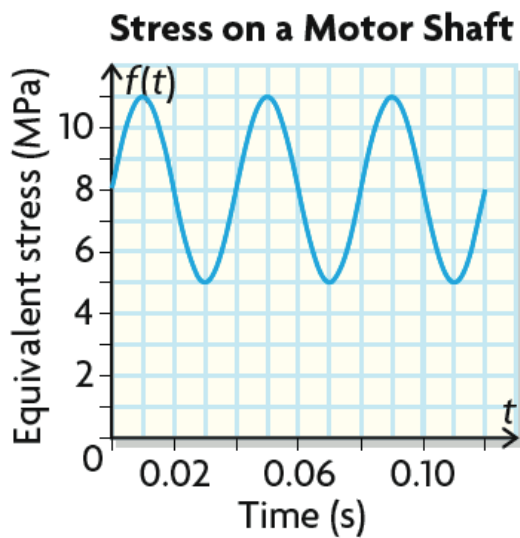
a) Represent the function with an equation using a sine function

b) Represent the function with an equation using a cosine function

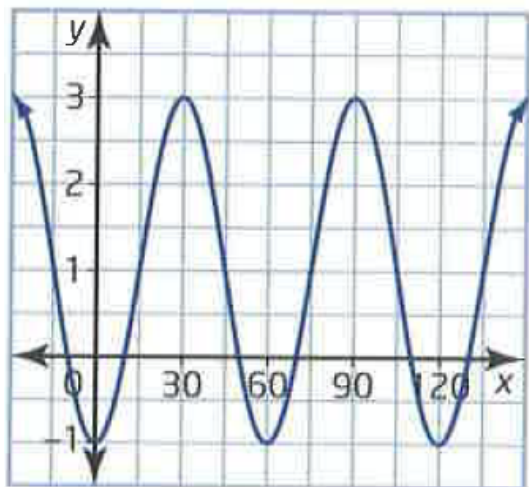
3) Determine the equation of a cosine function that represents the graph shown.



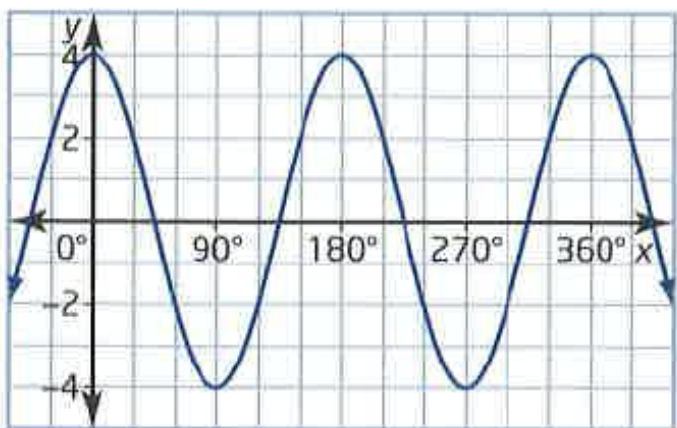
4) The relationship between the stress on the shaft of an electric motor and time can be modelled with a sinusoidal function. Determine an equation of a function that describes stress in terms of time.



5) Determine the equation of the sine function shown.



6) Represent the graph of the following functions using a sine and cosine function.



Answers

1) **a)** $y = 5 \sin [3(x + 30^\circ)] - 2$ **b)** $y = 5 \cos 3x - 2$

2) **a)** $y = \frac{1}{2} \sin \left[\frac{1}{2}(x + 180^\circ) \right] + 1$ **b)** $y = \frac{1}{2} \cos \frac{1}{2}x + 1$

3) $y = \cos [4(x - 82.5^\circ)] + 2$

4) $y = 3 \sin (9000x) + 8$ OR $y = 3 \cos [9000(x - 0.01)] + 8$

5) **a)** $y = 2 \sin [6(x - 15^\circ)] + 1$

6) $y = 4 \cos 2x$ and $y = 4 \sin [2(x + 45^\circ)]$.